

Factors affecting parallel performance

ECM3446/ECMM461: High Performance Computing

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Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manager worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Performance degradation

SAB: section 1.4.5

- A number of factors can contribute to real-world parallel performance being less than the idealised peak performance:
 - **S**tarvation: insufficient parallel work to keep the processors busy (insufficient parallel work, uneven distribution of work)
 - **L**atency: time taken for information to travel from one part of the system to another (e.g. accessing memory, sending messages)
 - **O**verhead: work required in addition to the computation (e.g. starting and stopping OpenMP parallel regions)
 - **W**aiting: multiple threads/processes accessing a shared resource (e.g. contention for memory or network bandwidth)

Lecture Units

- Unit 7.1: Parallel scaling and parallel fraction
- Unit 7.2: Load balancing and loop scheduling
- Unit 7.3: Interconnect and messages
- Unit 7.4: Domain decomposition overhead
- Unit 7.5: Parallel overheads and Amdahl's law

Workshop 7

- In this workshop we will look at calculating the Mandelbrot set
- We will see how to load balance the calculation using OpenMP
- The workshop uses information from units 7.1 and 7.2